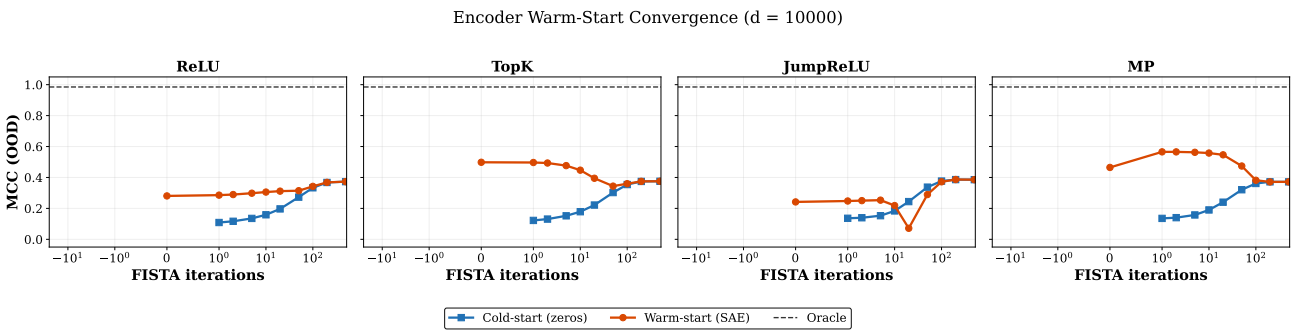

Figure 47: Encoder warm-start convergence, $d_z = 10000$.

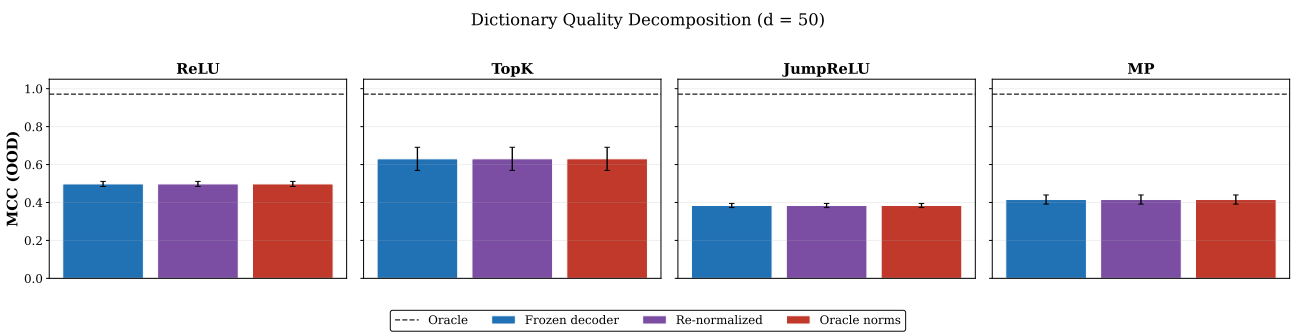

Figure 48: Dictionary quality decomposition, $d_z = 50$.

In Case 2, we compute the area (α_2) of the triangle between the classifier and the correct OOD decision boundary. The base between $\frac{1}{2}A_1$ and $\frac{1}{2}A_1 + A_3$ has length 1. The angle on top is $a_2 = \pi - \theta$ and the angle below is $b_2 = \phi + \theta - \pi$. Thus, using equation 17, the area is

$$\alpha_2 = \frac{\sin(\pi - \theta)\sin(\phi + \theta - \pi)}{2\sin(\phi)} = \frac{\sin(\theta)\sin(\phi + \theta - \pi)}{2\sin(\phi)} \tag{20}$$

The OOD accuracy will be 100% minus the proportion that the area occupies in the left plus right parallelogram (2α)

$$acc_2(\text{OOD}) = 1 - \frac{\alpha_2}{2\alpha}. \tag{21}$$

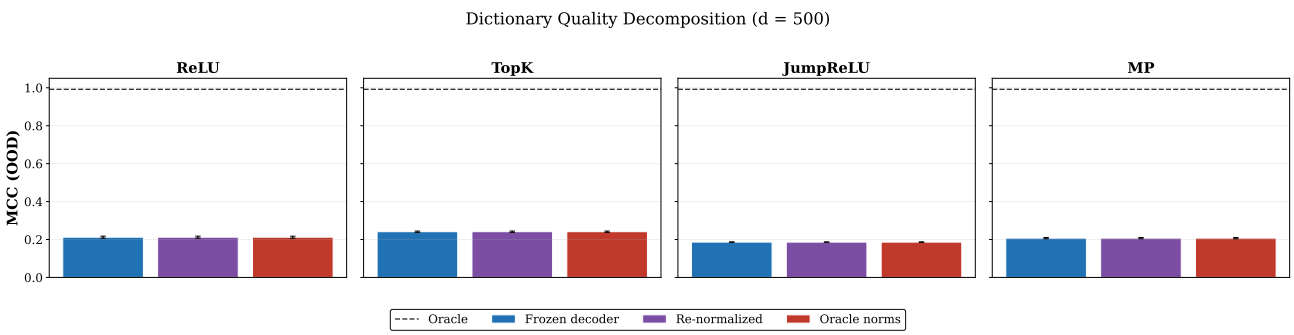

Figure 49: Dictionary quality decomposition, $d_z = 500$.

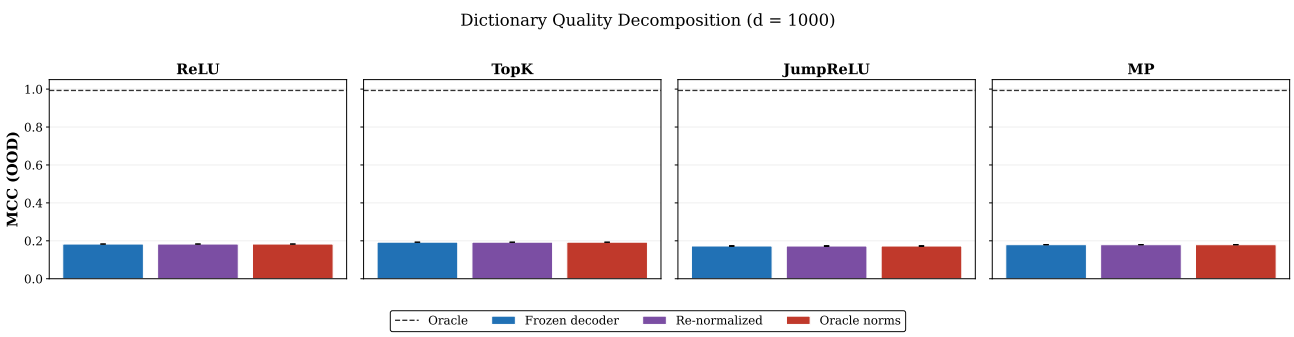

Figure 50: Dictionary quality decomposition, $d_z = 1000$.

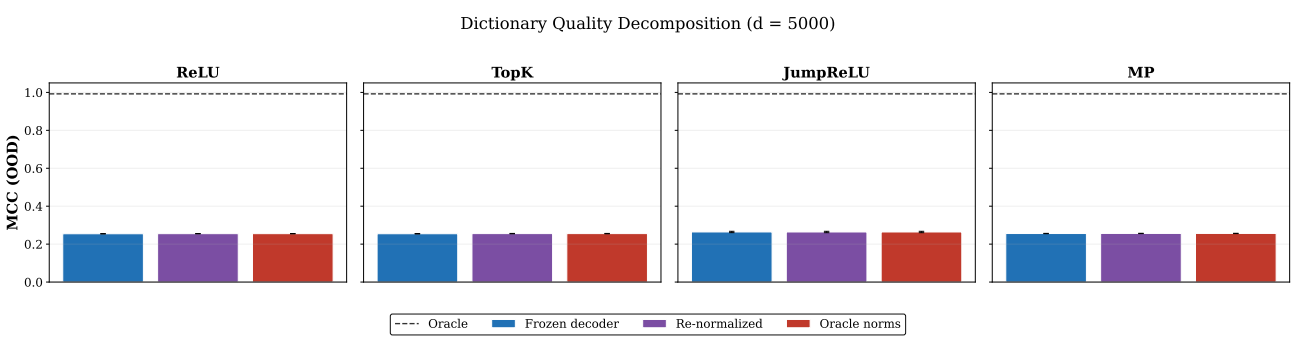

Figure 51: Dictionary quality decomposition, $d_z = 5000$.

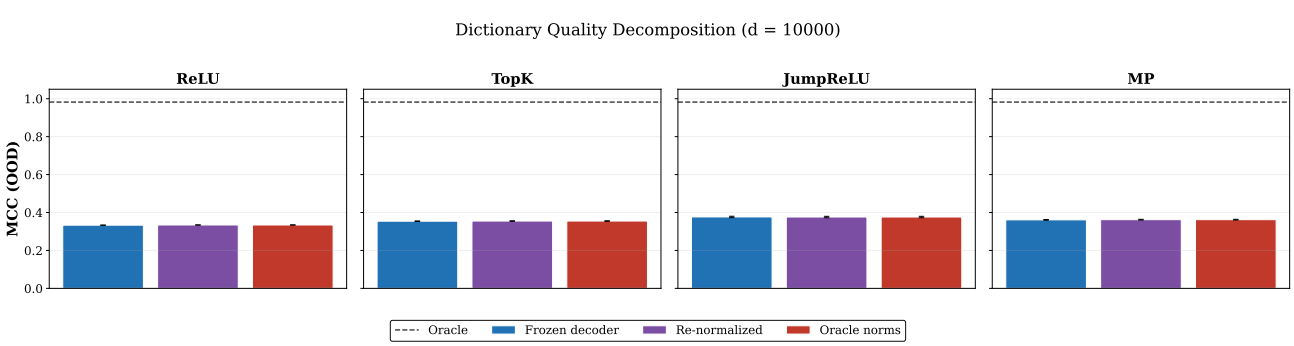

Figure 52: Dictionary quality decomposition, $d_z = 10000$.

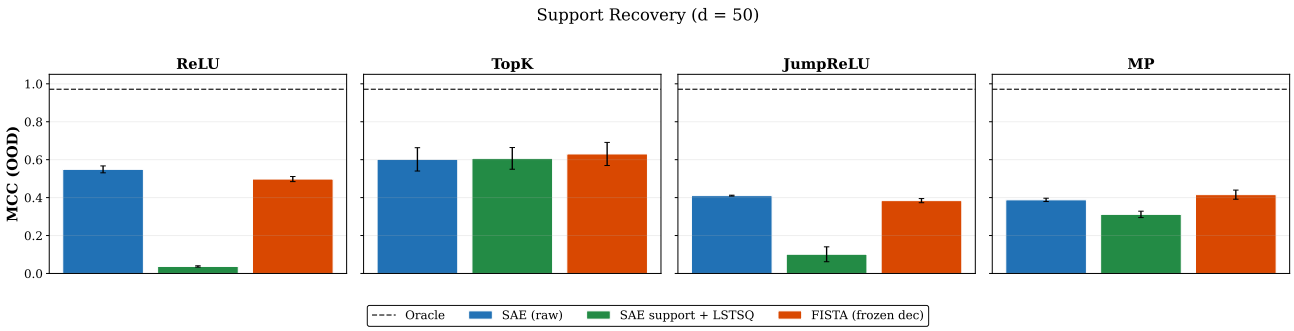

Figure 53: Support recovery, $d_z = 50$.

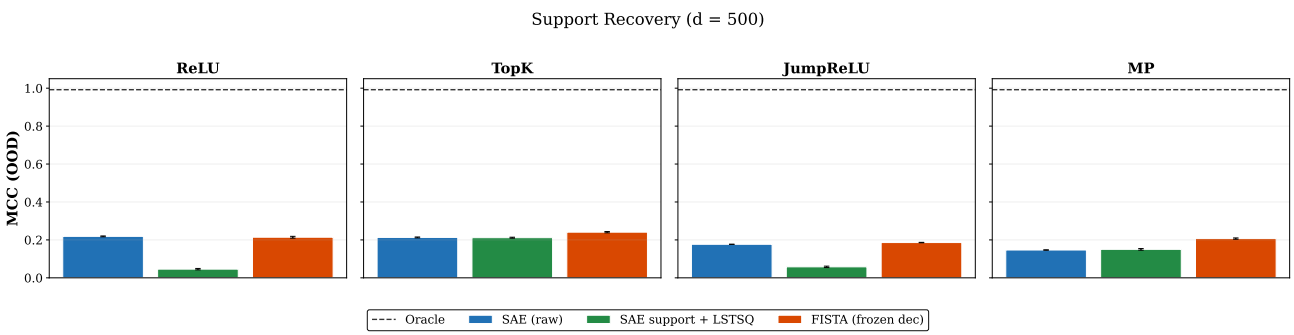

Figure 54: Support recovery, $d_z = 500$.

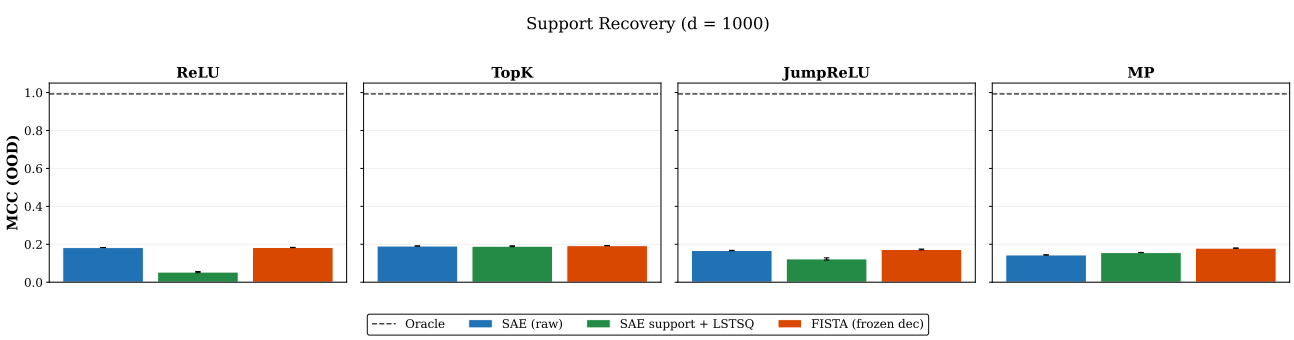

Figure 55: Support recovery, $d_z = 1000$.

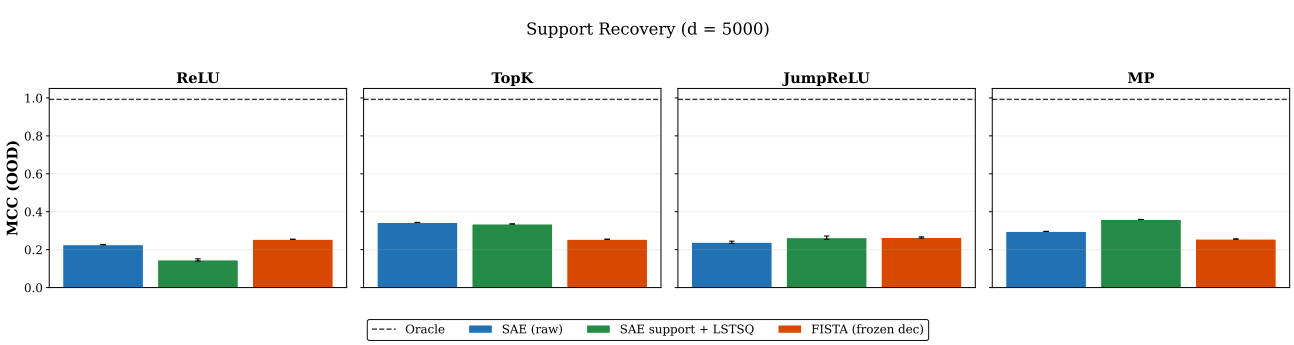

Figure 56: **Re-estimating magnitudes on the SAE's incorrect support degrades MCC.** Blue: raw SAE codes. Green: SAE support with least-squares magnitude re-estimation. Orange: FISTA on frozen decoder. Dashed: oracle. Only TopK's support is useful. $d_z = 5000$, $k = 10$.